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Executive Summary: Quantum computing utilizes qubits that can exist in quantum superpositions. This computational paradigm enables exponential acceleration for specific mathematical problems. Shor’s algorithm can factor large integers in polynomial time, threatening classical RSA encryption. Grover’s algorithm provides quadratic speedup for searching unstructured databases.

Post-Quantum Cryptography and Standards - Page 2

Mitigation Strategies: To counter quantum threats, NIST standardized post-quantum cryptography algorith

Primary lattice-based standards include ML-KEM for key encapsulation and ML-DSA for digital signatures.

Organizations are strongly advised to begin cryptographic inventory audits immediately.

Conclusion: Full migration to post-quantum cryptographic security is recommended before the year 2030.